

Linking Quality and Quantity of Parental Linguistic Input to Child Language Skills: A Meta-Analysis

Nina J. Anderson 
University of Denver

Susan A. Graham 
University of Calgary and Alberta Children's Hospital
Research Institute

Heather Prime
York University

Jennifer M. Jenkins
University of Toronto

Sheri Madigan 
University of Calgary and Alberta Children's Hospital
Research Institute

This meta-analysis examined associations between the quantity and quality of parental linguistic input and children's language. Pooled effect size for quality (i.e., vocabulary diversity and syntactic complexity; $k = 35$; $N = 1,958$; $r = .33$) was more robust than for quantity (i.e., number of words/tokens/utterances; $k = 33$; $N = 1,411$; $r = .20$) of linguistic input. For quality and quantity of parental linguistic input, effect sizes were stronger when input was observed in naturalistic contexts compared to free play tasks. For quality of parental linguistic input, effect sizes also increased as child age and observation length increased. Effect sizes were not moderated by socioeconomic status or child gender. Findings highlight parental linguistic input as a key environmental factor in children's language skills.

The development of early language skills is critical for success in numerous developmental domains. Early language ability has been linked to overall intellectual ability (Feldman et al., 2005), to cognitive abilities such as executive functioning (e.g., Wade, Browne, Madigan, Plamondon, & Jenkins, 2014), and to academic success (e.g., Agostin & Bain, 1997). Language skills also have demonstrated importance for socioemotional development in terms of emotion regulation (Hentges, Devereux, Graham, & Madigan, in press) and general social competence (e.g., Herbert-Myers, Guttentag, Swank, Smith, & Landry, 2006; Roben, Cole, & Armstrong, 2013), and they predict long-term functional outcomes including educational attainment and occupational achievement (Johnson, Beitchman, &

Brownlie, 2010). Although an individual's language ability is influenced by genetics (Stromswold, 2001), research suggests that environmental variables play a prominent role (Hoff, 2006). In the current meta-analysis, we examine one factor in children's linguistic environments, namely the quality and quantity of parental linguistic input.

To date, considerable research has examined the association between parenting and children's language development, including both the affective-relational environment (e.g., parental warmth and sensitive responsiveness), as well as the linguistic environment (e.g., quality and quantity of caregiver language input). With respect to the affective-relational environment, a recent meta-analysis by Madigan et al. (2019) demonstrated moderately strong associations between maternal sensitive responsiveness ($d = .55$; $N = 36$) and child language, as well as between maternal warmth and child language ($d = .30$; $N = 13$). A comprehensive synthesis of the literature examining the association between the

Funding: Social Sciences and Humanities Research Council of Canada. Alberta Children's Hospital Foundation through the Owerko Centre. Canada Research Chairs Program.

The authors would like to thank Cheri Nickel, MLIS, for her assistance with the search strategy.

Correspondence concerning this article should be addressed to Sheri Madigan, Department of Psychology, University of Calgary, 2500 University Dr. N.W., Calgary, Alberta, T2N 1N4, Canada. Electronic mail may be sent to sheri.madigan@ucalgary.ca.

© 2021 Society for Research in Child Development
All rights reserved. 0009-3920/2021/9202-0003
DOI: 10.1111/cdev.13508

parental linguistic environment and children's language skills, however, has yet to be conducted. In the current meta-analysis, we sought to characterize the relation between parental language input and children's language skills, and to assess potential moderators of this association.

Linguistic input is only one feature of children's broader language environment and there is evidence that parental linguistic input may be a limited indicator for certain groups, including lower socioeconomic status (SES) families (e.g., Sperry, Sperry, & Miller, 2019a, 2019b). Furthermore, there is ample evidence that children learn in contexts in which they receive indirect language exposure (e.g., Akhtar, Jipson, & Callanan, 2001; Akhtar, Tolins, & Tree, 2019). That said, there is considerable historical and current interest in, and emphasis on, this specific metric of the language environment (e.g., Hart & Risley, 1995; Vernon-Feagans, Bratsch-Hines, Reynolds, & Willoughby, 2019). As such, syntheses of the association between quantity and quality of direct parental linguistic input and children's language skills are relevant for our understanding of the role that this aspect of the language environment can play in potentiating children's language skills.

Parental Linguistic Input: Concept Definition and Empirical Support

Language input is typically described in terms of both its quantity and quality. *Quantity* of input is more consistently defined, and can be operationalized in terms of the number of words or tokens (Rowe, 2012) or number of utterances (Laks, Beckwith, Cohen, & Cohen, 1990) spoken to the child. In contrast, *quality* of input can refer to a number of different facets of language. As delineated in a recent review from Rowe and Snow (2020), quality of input can be analyzed along three dimensions: interactive features (e.g., responsiveness, joint-attention), linguistic features (e.g., clear pronunciation, lexical diversity, grammatical complexity), and conceptual features (e.g., contextualized and decontextualized topics). With respect to linguistic features, the diversity and complexity of spoken words is a foundational metric that is often considered in concert with the quantity of words spoken. This component of linguistic *quality* has been defined in terms of the use of different words or word roots (Baker, Vernon-Feagans, & The Family Life Project Investigators, 2015), type/token ratios (Bingham, Kwon, & Jeon, 2013), lexical richness (e.g., complexity and variety of words; Mol, & Neuman, 2014),

and rare word use (Beals & Tabors, 1993), among others.

In this meta-analysis, we included measures of linguistic input quality, measured by vocabulary diversity and syntactic complexity (i.e., different words, type/token ratios, etc.), and quantity (i.e., number of words, tokens or utterances) of language spoken by parents to their children. Given the breadth and heterogeneity of possible indices of input quality, we restricted our focus to vocabulary diversity and syntactic complexity in an effort to index a more uniform construct. Limiting our dimensions of interest enabled us to maintain a consistent approach to measuring parental linguistic input.

Research has generally found support for the notion that the quantity (e.g., Hart & Risley, 1995) and quality (e.g., Rowe, 2012) of linguistic input are correlated with child language outcomes, however, the strength of this association varies across studies. The magnitude of the association between parent linguistic input and child language may vary depending on the aspect of parent language under investigation, with quality of input potentially predicting child language more strongly than quantity of input (e.g., Hsu, Hadley, & Rispoli, 2017). This differential prediction may be dependent upon the age, and therefore language abilities, of the child (see Golinkoff, Hoff, Rowe, Tamis-LeMonda, & Hirsh-Pasek, 2019 for a summary). As such, the strength of the relation between parental linguistic input and child language skills remains unclear, as does the extent to which this association varies based on the type of input being examined. A meta-analysis helps to resolve discrepancies and yields a more precise estimate of the association between these constructs, that can in turn inform research and practice.

Potential Moderators of the Association Between Parent Linguistic Input and Child Language Skills

One central advantage to conducting meta-analyses is to assess potential moderators that may explain existing variation in effect sizes. In this meta-analysis, several sample, demographic, and measurement variables were considered as potential moderators.

Child Gender

Research suggests that girls may exhibit stronger receptive and expressive language skills in early development, but this difference declines between

20 and 24 months and is minimal thereafter (Huttenlocher, Haight, Bryk, Seltzer, & Lyons, 1991; Laks et al., 1990). Leaper and Smith (2004) found that girls were slightly more talkative than boys, but the overall effect size was negligible and was significant only in the youngest group of included children (12–35 months). Similarly, Fenson et al. (1994) found that gender accounted for only 1.9% of the variance in word production between 16 and 30 months of age.

The source of potential gender differences has been debated in the literature, and has included both biological and environmental explanations. Huttenlocher et al. (1991) argued that the gap reflects a difference in capacity for early vocabulary development, as they did not find evidence of differences in environmental stimulation measured by the frequency with which mothers spoke to sons versus daughters. However, meta-analytic evidence has suggested that mothers demonstrate a tendency to talk more with daughters than sons (Leaper, Anderson, & Sanders, 1998). Accordingly, in the current meta-analysis, child gender is examined as a potential source of between-study variability for the association between parental linguistic input and child language skills.

Parent Gender

Mothers and fathers can differ in their speech patterns, and thus, parent gender may also explain between-study variability. In terms of linguistic input quantity, a meta-analysis found that mothers were more talkative than fathers, though the effect size was small (Leaper et al., 1998). In terms of quality, there appears to be little overall difference between mothers' and fathers' speech diversity (Bernstein Ratner, 1988); however, findings examining differences in complexity are mixed (see Leech, Salo, Rowe, & Cabrera, 2013 for a summary). Thus, in the current meta-analysis parent gender will be examined as a potential moderator of the strength of the relation between parental linguistic input and child language.

Child Age

Two important age considerations that may explain variation in effect sizes across studies include the age of the child at the time parental linguistic input is assessed, and the age of the child when their language skills are assessed. There is evidence to suggest that the importance of different aspects of linguistic input may vary with

developmental stage. For example, Rowe (2012) demonstrated that quantity of input is most important for children's language skills during the second year of life, while diversity and complexity are more important during the third year. As such, we will examine whether the strength of the association between parental linguistic input and child language changes based on the age of the child.

Socioeconomic Status

Recent work has sparked a resurgence in interest in the controversial "30-million-word gap" (Hart & Risley, 1995), the notion that children from lower SES families in North America hear 30 million fewer words than those from middle class families by the time they are 4 years old. Sperry et al. (2019a) found evidence that contrasted with Hart and Risley's (1995) original findings, demonstrating that caregiver talkativeness varied by community, rather than SES, and that the "word gap" essentially disappeared when a more inclusive definition of the language environment was employed (e.g., including speech from all caregivers and/or all ambient speech that a child overhears). Though the magnitude (or even existence) of a contextually driven word-gap remains contentious (Golinkoff et al., 2019; Purpura, 2019; Sperry et al., 2019b), several studies do support this notion. For example, the speech of high-SES mothers has been demonstrated to be more complex, more diverse, and contain more words as compared to the speech of mothers from low/middle-SES backgrounds (Hoff, 2003; Huttenlocher, Vasilyeva, Waterfall, Vevea, & Hedges, 2007). In addition, Hoff (2013) demonstrated that children from lower SES homes show deficits in their language trajectories, and that a significant source of these deficits is derived from differences in their language experiences. Given that environmental influences can be more important in lower versus higher SES groups for some cognitive abilities (e.g., Madigan et al., 2019; Turkheimer, Haley, Waldron, d'Onofrio, & Gottesman, 2003), the relation between linguistic input and child language may be stronger in lower SES groups. Such a finding would be in line with a risk-resilience framework, wherein high-quality linguistic input serves as a protective factor in the context of risk.

Measurement

The diversity of measurement techniques available to assess parent and child language suggests that measurement factors may explain some of the

heterogeneity between studies in this area. As such, measurement factors will be examined as potential moderators of the association between parental linguistic input and child language in the current meta-analysis.

Assessment type. The manner in which parent and child language are measured can create variation in effect sizes across study findings. Children's language ability is assessed through a diverse set of methodologies, including structured assessments, parental questionnaires, and observational data. Disagreement exists regarding the optimal methods for assessing child language. For example, some have argued that standardized measures are preferable to observing child language during concurrent parent-child interactions, as child language during these interactions may be contaminated by the amount of parental speech (Rowe, 2012). It has also been argued that observations of spontaneous speech cannot be considered a "pure" test of vocabulary, as it is also a factor of style of language use (Hoff, 2003). That is, a shy or otherwise less talkative child may underperform in observational measures of spontaneous speech. On the other hand, some have argued that observational assessments are more ecologically valid measures of child language as compared to lab assessments (see Bornstein & Haynes, 1998 for a comparison of observation, assessment, and maternal report measures). Finally, parental reports are believed to overcome some of the restrictions associated with observation and standardized assessments, as they are less bound to a specific context and are not dependent on a child's cooperation in a lab task (Bornstein & Haynes, 1998). Additionally, parents typically spend a great deal of time with their children and know them well, potentially allowing for more accurate reporting. On the other hand, these measures may be subject to various reporting biases. Based on these arguments, there is no one measure type that is clearly superior to all others across circumstances (Bornstein & Haynes, 1998).

Consistent with the lack of clear support for any one measurement type using theoretically based arguments, research has found that parental reports show significant correspondence with lab-based measures, including observational language samples and structured assessments (Bornstein & Haynes, 1998; Fenson et al., 1994 for concurrent validity specific to the MacArthur Communicative Development Inventories). As such, these findings suggest that all three measurement types may yield similar estimates of child language and may be similarly valid. Given the breadth of measures available to

assess child language, however, we sought to address the measurement debate by examining whether effect sizes are moderated by different methods of child language assessment.

Observation is the principal method of assessing parent linguistic input; however, the specific conditions often vary widely from study to study, including location, context type, and length of observation, and this may contribute to differences across studies. For example, parents may speak more to their children when in the lab due to social desirability, or may speak more at home due to feeling more at ease. As such, observation location may be a moderator of the association between parent language input and child language. In terms of context type, findings in this area are mixed and have mainly focused on comparing free-play observations to naturalistic observation during meal times. Although Bornstein, Tamis-LeMonda, and Haynes (1999) found that older children and mothers used more words during play than during meal times, Beals and Tabors (1993) found that the number of tokens evoked during playtime and mealtime were similar, but mealtime generated more word types. Finally, longer observations may yield more representative data, as they are more likely to capture the parent and child's language in various contexts. Given that longer observations can be methodologically difficult to execute, examining whether the association between linguistic input and child language differs based on the observation length will inform future research efforts on assessment periods.

Current Study

Previous meta-analyses have examined the effects of parent-implemented language interventions on child language outcomes (Law, Garrett, & Nye, 2004; Roberts, Curtis, Sone, & Hampton, 2019; Roberts & Kaiser, 2011), however, these studies were based on intervention studies exclusively, rather than observational studies. In this meta-analysis, the primary aim was to derive pooled estimates and compare the magnitude of associations between quality and quantity of parental linguistic input, respectively, and child language in observational studies. Given that the existence of an association between parental linguistic input and child language has already been established in the literature (e.g., Hart & Risley, 1995; Rowe, 2012), our analyses seeking to characterize the magnitude of the association between these variables were confirmatory in nature. Given recent interest in exploring

the impact of different linguistic features on children's language across developmental stages (e.g., Golinkoff et al., 2019; Jones & Rowland, 2017; Rowe, 2012), particularly with respect to input quality (Rowe & Snow, 2020), we will also examine whether the effect sizes of our two measures of input quality, diversity and complexity, differ in magnitude, as well as their relation to child language at different ages (Rowe, 2012). These analyses were exploratory in that no prior work has specifically suggested that diversity and complexity are differentially associated with child language across development. However, these analyses will contribute to the ongoing efforts to understand what constitutes optimal linguistic input at different points in development.

A secondary aim was to identify potential moderators of the associations between quality and quantity of parental linguistic input and child language to gain a comprehensive understanding of the characteristics that influence the strength of these relations. Potential moderators include sample (child age, child gender, parent gender), demographic (SES), and measurement characteristics (measurement type, observation details). We also examine study design (cross-sectional vs. longitudinal) and dissemination medium (published versus unpublished) as is typical in meta-analytic research in developmental science (e.g., Quinn, Donnelly, & Kidd, 2018). The moderator analyses are exploratory, given that previous research has suggested, but not conclusively demonstrated, that these variables may amplify or attenuate associations between parental linguistic input and child language outcomes.

Method

Definitional Criteria

Quantity of input was defined as the amount of speech directed to the child by the parent, including the number of words/tokens and number of utterances. *Quality* of input included the diversity and complexity of the parents' speech, with diversity including measures such as the number of different words or types of words used, and complexity including measures such as mean length of utterance (MLU; a measure of the complexity of sentence structure, typically indexed by the number of morphemes; Crystal, 2008), rare word use, and lexical richness (i.e., complexity and variety of words). Note that, while we include MLU as a measure of quality, we recognize that MLU can be considered both a measure of quantity and a measure of

complexity (i.e., quality). In reviewing the articles that measured MLU in this meta-analysis, nine articles explicitly refer to it as a complexity measure, two describe it as a measure of quantity, two refer to it as both quantity and complexity, and the remaining seven do not specify a classification. Our goal was to define our constructs in a manner consistent with and representative of the current literature, and thus we categorized MLU as a complexity (and therefore quality) measure.

Child language skill was assessed in terms of expressive and receptive language. Expressive language was defined as children's productive speech, including measures of lexical richness, language productivity, expressive vocabulary diversity, etc. Receptive language was defined as children's understanding of spoken language/vocabulary.

Search Strategy

A medical librarian conducted searches in PsycINFO, MEDLINE, EMBASE, Web of Science, and Dissertation Abstracts International for published and unpublished studies up to July 2017. The search was restricted to studies written in English. Database-specific headings and text word fields were searched for concepts of "parent," "child," and "language," with truncation symbols used to capture variant endings and spellings. Synonymous terms were combined with the Boolean "OR", and the parent, child, and language concepts were combined with the Boolean "AND". The detailed search strategy can be found in Table S1, and the PRISMA flow diagram can be found in Figure 1. Additionally, the reference lists of all included articles were screened for additional relevant studies, which resulted in the addition of two studies not caught by the search strategy.

Inclusion and Exclusion Criteria

Studies were screened by two independent coders for the following inclusion criteria: (a) there was an assessment of the quantity and/or quality of parent linguistic input; (b) there was an assessment of child expressive or receptive language; (c) language was assessed in English; (d) the study reported a statistic that allowed for the calculation of an effect size. When titles and abstracts were insufficient to determine eligibility criteria, full texts were retrieved. A summary of the inclusion criteria used is in Table S2.

Exclusion criteria were as follows: measures examining specific types/purposes of verbal input

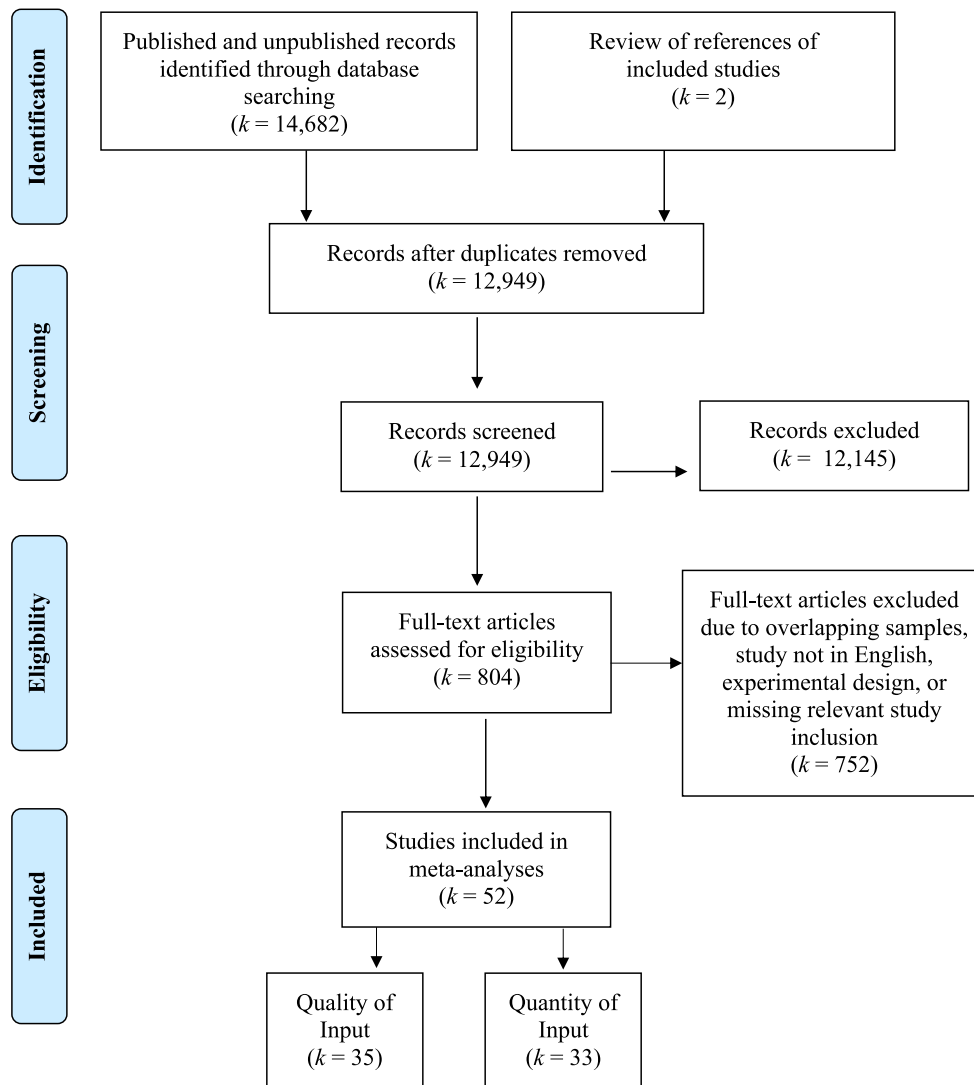


Figure 1. PRISMA flow used to identify studies for detailed analysis of parenting and children's language. [Color figure can be viewed at wileyonlinelibrary.com]

(e.g., repetitions, descriptions, labels, questions, decontextualized language, demands), or aspects of vocal prosody (e.g., pitch, length of sounds, volume diversity). These aspects of communication were beyond the scope of linguistic input that we sought to assess, which focused on the quantity and quality of word- and utterance-level speech that parents directed towards their children. These other factors were sufficiently diverse that they presented a challenge to classify and were thus counter to our goals of maintaining a uniform definition. Precursors to verbal abilities and non-verbal abilities, such as babbling and gesturing, were not included. Composite measures that combined language with other abilities (e.g., cognitive ability, reading, writing) were excluded. Atypically developing samples,

including children with language delays, intellectual disabilities, deafness, etc., were excluded. Longitudinal studies were not included if the child measure was taken at an earlier time point than the parent measure, as this did not address our research question. Intervention studies were included if they reported a pretest/baseline estimate of both parent and child language. A total of 52 studies met our criteria and were included in one or both meta-analyses ($k = 35$ for *quality* of linguistic input; $k = 33$ for *quantity* of linguistic input).

Study Variable Coding

The first author extracted all effect size data and potential moderator variables from the 52 studies

meeting inclusion criteria. A second coder performed data extraction on 10% of studies to determine inter-rater reliability. Percent agreement for categorical moderators was 91%, for continuous moderators agreement was 0.90, and agreement on the extraction of effect size data was 100%. Any discrepancies were resolved by review and discussion, and consensus coding was used in data analysis. Both adjusted and unadjusted effect sizes were extracted, and if both were provided we selected the unadjusted effects, which were the most frequently reported across studies. Categorical moderators were determined based on 80% or more of the sample falling in a given category. If information for any given moderator was missing, it was coded as "not specified," unless indicated below.

Sample Characteristics

The percentage of female children and male parents were coded. Consistent with other studies, if no information was provided regarding the gender distribution of the child sample, gender was coded as 50% female ($k = 1$). Child age in months was coded at the time of the parent linguistic input measure and the child language measure.

Demographic Characteristics

SES was categorized as (a) middle/upper, (b) low, or (c) diverse. These classifications were typically based on how samples were defined by the authors (e.g., a low income sample, a middle to high SES sample, etc.). However, if authors did not provide a classification, SES was determined based on the reported income and education of participants. Given that race/ethnicity and SES are often confounded in studies examining contextually driven word-gaps, we focus only on the potential moderating role of SES, rather than race/ethnicity, in our analyses (Vernon-Feagans et al., 2019). Demographic information including the percentage of participants belonging to minority race/ethnic groups in each sample is provided in Table S3.

Child Language

Child language was categorized as (a) receptive, (b) expressive, or (c) combined expressive and receptive language. The method for assessing child language was coded as being a(n) (a) questionnaire, (b) observation, (c) standardized assessment, or (d) combination of measures. Finally, the informant for the child language measure was categorized as the

(a) parent, (b) teacher, (c) trained coder, (d) home visitor/experimenter, or v) a combination.

Parent Linguistic Input

Parent linguistic input was categorized as (a) quantity and/or (b) quality, with measures of quality including (a) diversity, (b) complexity, or (c) combined measures. Parent linguistic input was assessed using observational data across all included studies. The location of the observation was coded as taking place in the (a) home, (b) laboratory, or (c) a combination of the home and laboratory. The context of the observation was coded as occurring during (a) free play (i.e., parents were instructed to play with children as they normally would, with no restrictions), (b) a semi-structured task (e.g., shared book reading), (c) naturalistic/home observations (i.e., parents and children going about their daily lives, including meal time observations, morning routine observations, etc.), or (d) a combination of these contexts. The length of the observation was coded in minutes.

Study Design and Publication Characteristics

The study was classified as either (a) longitudinal or (b) cross-sectional. The dissemination medium (i.e., journal vs. dissertation) was also extracted.

Effect Size Extraction

Two data sets were created to organize the extracted data, in order to allow for the examination of both quality and quantity of linguistic input, independently. Each sample was only represented once within each data set.

Overlapping Samples

Two steps were taken to ensure that only independent effect sizes were represented within each meta-analysis. First, we extracted any notable characteristics pertaining to sample and recruitment, including if the sample was part of a longitudinal/resource study (e.g., Family Life Project, NICHD Study of Early Child Care and Youth Development). Next, we cross-referenced authors across studies, using first and last authors as a reference point. If sample overlap was identified, the study with the largest sample size and most comprehensive information available for data extraction was selected for inclusion.

Multiple Effect Sizes From a Given Study

Studies often reported multiple statistics that were relevant for calculating an effect size. Several steps were taken to ensure that only one effect size was represented for each study. First, if the parent and child measures were taken at multiple time points, we selected the statistic with the longest time between the parent and child assessments (i.e., longitudinal over cross-sectional). Second, if the time between all measures was equal, we selected the language measure taken when the child was the oldest, in order to capture more developed language abilities ($k = 2$). After implementing these criteria, some studies still contained multiple relevant statistics if they employed multiple measurement types. To address this, we created a measurement hierarchy to determine our selection of statistics, based on the relative frequency of each measure reported in our included studies. By creating such a hierarchy, we hoped to represent measurement techniques most typically used in this area of research. This method of statistic selection has been used in other language meta-analyses (Roberts & Kaiser, 2011). Based on our hierarchy, we selected more frequently used measures (e.g., word types) over infrequently used measures (e.g., word roots). Overall, the measure most often used to assess linguistic input quantity was number of words/tokens, while quality was most often assessed using MLU for complexity and the number of types for diversity (note that if a study provided measures of both complexity and diversity, the statistics were pooled for the input quality meta-analysis, but associations between complexity and child language, as well as diversity and language, were examined independently in a sub-analysis). For child language, the Peabody Picture Vocabulary Test was used most often to assess child receptive language, and MLU and number of different words/types was reported most frequently for expressive language. If a single study provided a measure of receptive and expressive language, these two effect sizes were pooled to provide the most global and representative assessment of child language. A summary of the criteria and processes used to determine the selection/inclusion of statistics is reported in Table S2.

Data Preparation and Analyses

Extracted data were entered into Comprehensive Meta-Analysis software (CMA, version 3.0; Borenstein, Rothstein, & Cohen, 2005), which was used to calculate and analyze effect sizes and to conduct

moderator analyses. Effect size calculations (r) were computed with 95% confidence intervals around the pooled estimate. Effect sizes were weighted based on their inverse variance, thus giving greater weight to studies with larger samples and more precise effect sizes (Hedges & Olkin, 1985). Similar to other meta-analyses, studies that reported non-significant effect sizes ($k = 7$) without any additional statistical information were imputed as a p -value of .50 (Rosenthal, 1995). Random effects models were used for calculations, which considers each study to have its own population parameters and measurement methods and thus, more accurately reflects between-study heterogeneity (Rosenthal, 1995). To test for statistical differences between pooled effect sizes for partially overlapping studies, as some studies reported both quality and quantity of input, as well as diversity and complexity of input, we computed 85% confidence intervals around the point estimates as a conservative significance test (Goldstein & Healy, 1995), consistent with recommendations by Julious (2004) and other meta-analyses in child development (e.g., Fearon, Bakermans-Kranenburg, van IJzendoorn, Lapsley, & Roisman, 2010). Q -statistics were used to determine heterogeneity of effect size and significance of categorical moderators (Borenstein, Hedges, Higgins, & Rothstein, 2009). Continuous moderators were examined using mixed effects meta-regressions (Thompson & Higgins, 2002). Publication bias was assessed using the trim and fill procedure (Duval & Tweedie, 2000), including examination of funnel plots. If the funnel plot exhibits asymmetry, such that fewer studies fall along the bottom left of the mean effect size, the trim and fill procedure imputes symmetrical extreme values to balance the funnel plot and yields an adjusted mean effect size accounting for the publication bias.

Results

Study Characteristics

Detailed sample and study characteristics are shown in Table S4. The sample sizes ranged from 5 to 551 (total $N = 2,754$; mean $N = 53$). At the time of the parent and child language assessments, respectively, children were on average 28.56 months old (range = 1–67 months, $SD = 15.76$ months) and 33.69 months old (range = 12–70 months, $SD = 15.71$ months).

Of note, several seminal studies could not be included in the current synthesis due to the statistical methods employed by the individual studies. Specifically, studies that presented findings in the

form of individual growth modelling of longitudinal data were not included. Authors of these studies were contacted in an effort to obtain raw data. However, data were unavailable for several reasons, which are summarized in Table S5.

Meta-Analysis of Quality of Parental Linguistic Input and Child Language

Effect Size Estimate

The pooled effect size for the association between quality of linguistic input and child language across 35 samples (pooled $N = 1,958$) was significant

($r = .33$; $p < .001$; 95% CI [.24, .42]; see Figure 2). Examination of the funnel plot (see Figure S1) revealed no evidence of publication bias. The Q statistic was significant ($Q = 122.30$; $p < .001$), indicating heterogeneity of effect sizes. Moderator analyses were thus conducted to explain this variability, including demographic, measurement, and publication moderators (see Table 1). Significant moderators are described below.

Moderator Analyses

Effect sizes varied by age of child at the time of the parental linguistic input ($k = 35$; $b = .009$;

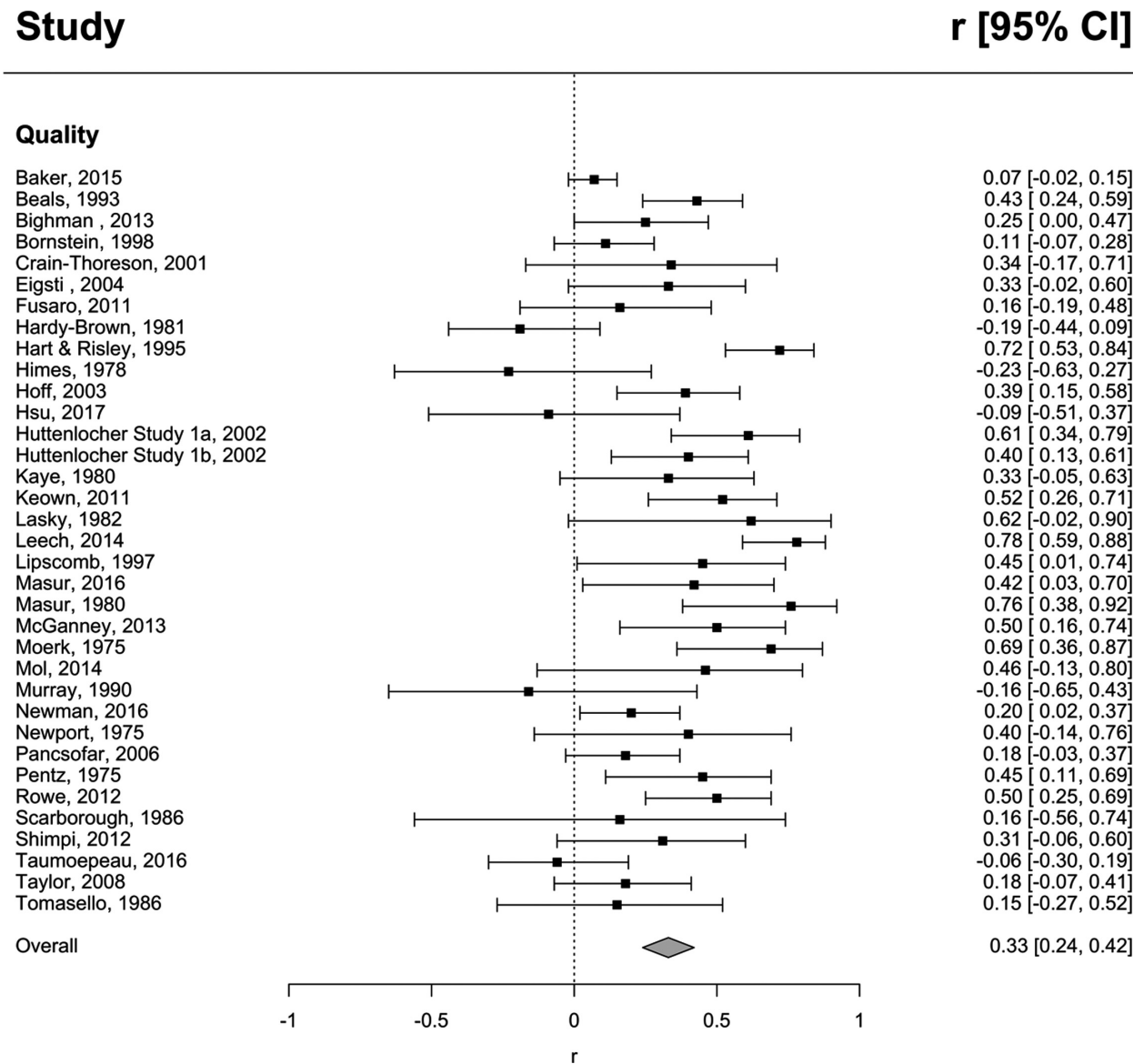


Figure 2. Forest plot of the effect sizes for each study included in the meta-analysis on quality of linguistic input and child language. Observed effect sizes and 95% CIs are indicated for each sample.

Table 1
Results of Moderators for the Association Between Quality of Linguistic Input and Child Language

Categorical moderators	<i>k</i>	<i>r</i>	95% CI	Homogeneity <i>Q</i>	<i>p</i> -value
Socio-economic status				0.63	.73
Low	3	.39***	.34, .52		
Middle to upper	20	.31***	.19, .43		
Diverse	12	.36***	.17, .52		
Parent observation location				0.38	.54
Home	20	.36***	.24, .46		
Research laboratory	14	.29**	.12, .45		
Parent observation context				14.13	.01
Free play	15	.26**	.15, .35		
Structured task	5	.29	-.01, .54		
Naturalistic observation	5	.58***	.45, .69		
Combination	10	.28**	.12, .43		
Language type				0.05	.98
Receptive	5	.31*	.07, .52		
Expressive	23	.34***	.22, .45		
General language	7	.33***	.15, .49		
Language method				7.27	.07
Questionnaire	5	.20	-.03, .41		
Observation	16	.47***	.32, .59		
Direct assessment	11	.27***	.14, .39		
Combination	3	.13	-.21, .44		
Language informant				7.17	.07
Parent	5	.20	-.03, .41		
Trained coder	18	.45***	.32, .55		
Experimenter	8	.25**	.09, .40		
Combination	4	.17	-.10, .42		
Dissemination medium				0.34	.56
Dissertation	6	.38***	.21, .53		
Publication	29	.33***	.22, .42		
Study design				2.18	.14
Cross-sectional	20	.39***	.26, .51		
Longitudinal	15	.26***	.13, .38		
Continuous moderator	<i>k</i>	<i>b</i>	95% CI	Z-value	<i>p</i> -value
Child age at parent measure	35	.009	.002, .015	2.73	.01
Child age at language measure	35	.007	.001, .013	2.15	.04
Percent of females in sample	35	.005	-.003, .013	1.14	.26
Percent of male parents	35	.001	-.005, .006	0.20	.85
Observation length ^a	28	.003	.000, .006	2.18	.03

^aThe Hart and Risley (1995) study was removed from this analysis as it was an outlier (3 + *SDs*) in terms of observation length.

* $p < .05$. ** $p < .01$. *** $p < .001$.

$p < .01$; 95% CI [.002, .015]) and child language measures ($k = 35$; $b = .007$; $p < .05$; 95% CI [.001, .013]). Specifically, larger effect sizes were found in studies assessing quality of parental linguistic input and child language at later ages. Larger effect sizes were also found when parent language was assessed in naturalistic settings ($k = 5$; $r = .58$; $p < .001$; 95% CI [.45, .69]) versus free play ($k = 15$;

$r = .26$; $p < .001$; 95% CI [.15, .35]), semi-structured tasks ($k = 5$; $r = .29$; $p < .06$; 95% CI [-.01, .54]), or a combination of tasks ($k = 10$; $r = .28$; $p < .01$; 95% CI [.12, .43]). Effect sizes also varied by observational length of linguistic input, with effect sizes increasing as the observation length increased ($k = 28$; $b = .003$; $p < .03$; CI [.000, .006]). No other significant moderators emerged.

Sub-Analyses of Diversity and Complexity of Input

Studies examining input quality included measures of both diversity and complexity. Given that age significantly moderated the association between linguistic input quality and child language, we then sought to examine whether either diversity or complexity was driving this effect, in order to better understand the relative role of these dimensions of linguistic quality across development. Confidence intervals (85%) around the pooled effect sizes for complexity ($k = 25$; $r = .33$; 85% CI [.26, .41]) and diversity ($k = 17$; $r = .27$; 85% CI [.18, .36]) of linguistic input and child language were used to assess statistical significance between the two pooled effect sizes. The 85% confidence intervals did overlap for these two pooled estimates, suggesting no significant differences in the effect size estimates for complexity and diversity of parent linguistic input. To further examine the pattern of associations however, we also examined whether each pooled effect size was moderated by child age. The associations between complexity of input and child language ($b = .004$; 95% CI [−.002, .011], $z = 1.28$, $p = .20$), and diversity of input ($b = .005$; 95% CI [−.003, .014], $z = 1.20$, $p = .23$) and child language were not moderated by child age. These results do not support the hypothesis that diversity and complexity of input individually matter relatively more or less at different ages.

*Meta-Analysis of Quantity of Linguistic Input and
Child Language*

Effect Size Estimate

A separate meta-analysis examining the pooled association between quantity of linguistic input and child language across 33 samples (pooled $N = 1,411$) was significant ($r = .20$; $p < .001$; 95% CI [.13, .27]; see Figure 3). There was indication of publication bias (see Figure S2). Duval and Tweedie's (2000) trim and fill procedure suggested two studies were missing to the left of the mean. With these studies imputed, the adjusted effect size was $r = .19$ (95% CI [.11, .26]). The Q statistic was significant ($Q = 53.98$; $p < .01$) and therefore moderator analyses were explored (see Table 2).

Similar to the meta-analysis on quality of input and child language, larger effect sizes were observed when parent language was assessed in naturalistic settings ($k = 6$; $r = .39$; $p < .001$; 95% CI [.23, .53]) versus free play ($k = 15$; $r = .19$; $p < .001$; 95% CI [.09, .29]) or a combination of tasks ($k = 7$;

$r = .09$; $p = ns$; 95% CI [−.05, .23]). It was also observed that effect sizes were stronger in longitudinal ($k = 17$; $r = .29$; $p < .001$; 95% CI [.17, .39]) versus cross sectional ($k = 16$; $r = .12$; $p < .01$; 95% CI [.04, .20]) studies.

*Comparison of Pooled Effect Sizes for Quality Versus
Quantity of Linguistic Input and Child Language*

Confidence intervals (85%) around the pooled effect sizes for quality ($k = 35$; $r = .33$; 85% CI [.26, .40]) and quantity ($k = 33$; $r = .20$; 85% CI [.15, .25]) of linguistic input and child language were used to assess statistical significance between the two effect sizes. As 85% confidence intervals do not overlap for these pooled estimates, they are considered to be significantly different and suggest stronger effects for parental linguistic quality and child language compared to quantity of linguistic input and child language.

Discussion

The current meta-analysis provides a pooled estimate of the association between parental linguistic input and children's language skills across the extant body of literature, indicating that there is a significant and moderate to large sized association (Funder & Ozer, 2019) between the quality (indexed by vocabulary diversity and syntactic complexity; $r = .33$) and quantity (indexed by number of words, tokens, or utterances; $r = .20$) of parent linguistic input and children's language development, with the effect size for input quality being more statistically robust than the effect size for quantity. There was significant between-study heterogeneity for the association between input quality and quantity and child language, and results suggests that the strength of the effects differ across development and measurement conditions.

Linguistic Quality Versus Quantity of Input

The effect sizes for the associations between the quantity and quality of parent linguistic input were significantly different from one another. Thus, while both of these aspects of linguistic input are important for child language, our findings show that quality of linguistic input is more strongly associated with child language outcomes than the quantity of linguistic input, with the effect of quality increasing as the child ages. Results are consistent with the proposal that different linguistic features

Study

r [95% CI]

Quantity

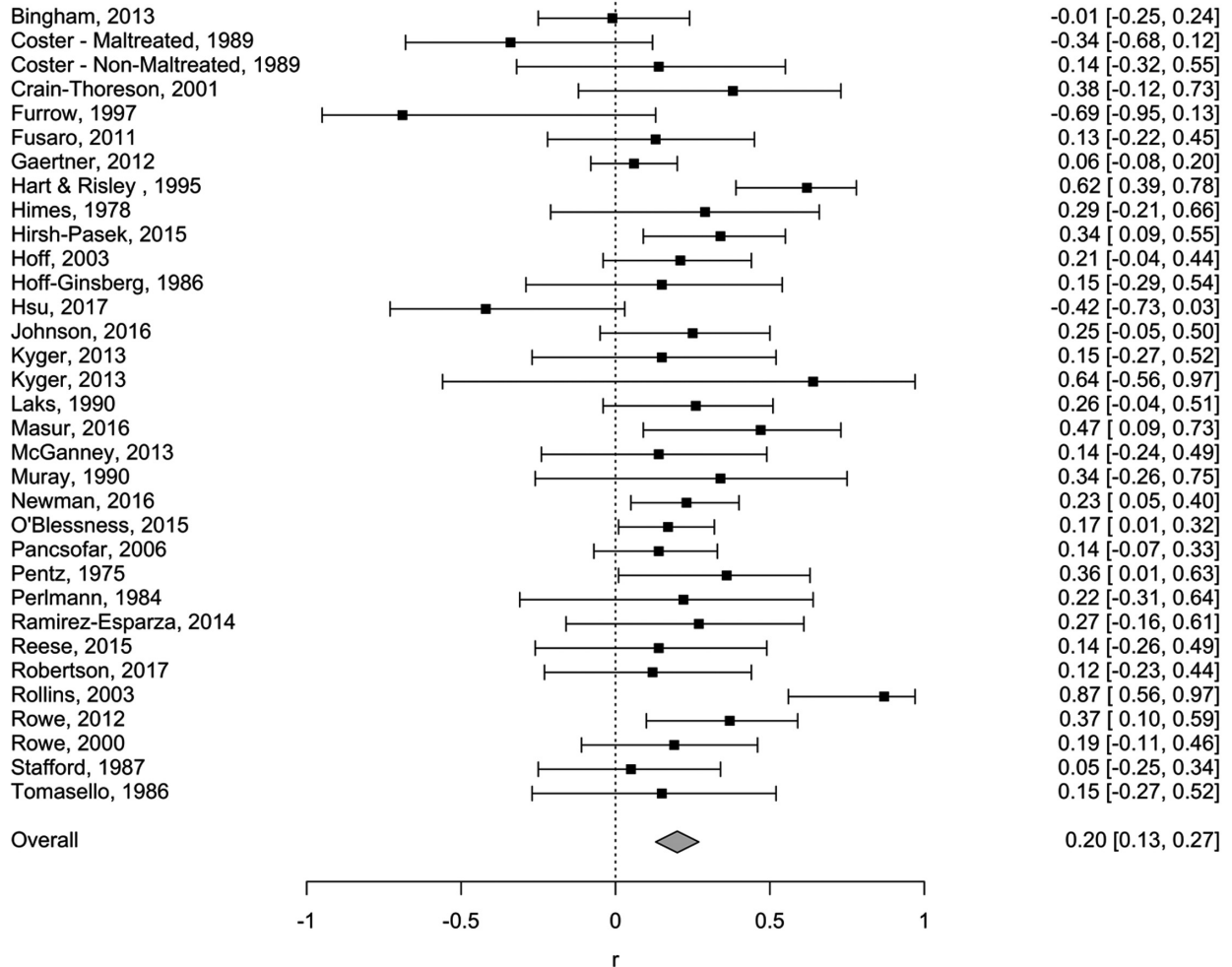


Figure 3. Forest plot of the effect sizes for each study included in the meta-analysis on quantity of linguistic input and child language. Observed effect sizes and 95% CIs are indicated for each sample.

pertaining to the quality of parents' speech may be relatively more important for children's language development beginning in toddlerhood than the sheer amount of parents' speech to which they are exposed (Golinkoff et al., 2019; Jones & Rowland, 2017; Rowe, 2012; Rowe & Snow, 2020). That is, quantity of language may help to initiate the language learning process, while input quality becomes more important once children have the foundational capacity to benefit from such speech (Golinkoff et al., 2019; Jones & Rowland, 2017; Rowe, 2012). However, it is important to note here that the quality measures examined in this meta-

analysis encompassed only diversity and complexity of speech. That is, using highly complex utterances may not be beneficial to the very young language learner and instead may be more relevant during the toddler and preschool years (Rowe & Snow, 2020). It is possible that other measures of input quality (i.e., verbal responsivity) would show a different relation with age.

To further explore whether the moderating effects of age varied by quality measure, the associations between the *diversity* ($k = 17$) and *complexity* ($k = 25$) of linguistic input and child language were examined separately to determine whether one of

Table 2

Results of Moderators for the Association Between Quantity of Linguistic Input and Child Language

Categorical moderators	<i>k</i>	<i>d</i>	95% CI	Homogeneity <i>Q</i>	<i>p</i> -value
Socio-economic status				3.50	.18
Low	5	.16	−.01, .32		
Middle to upper	23	.17***	.08, .26		
Diverse	5	.36***	.17, .53		
Parent observation location				0.61	.74
Home	17	.21***	.11, .31		
Research laboratory	15	.19**	.07, .30		
Parent observation context				7.80	.05
Free play	15	.19***	.09, .29		
Structured task	4	.22**	.06, .38		
Naturalistic observation	6	.39***	.23, .53		
Combination	7	.09	−.05, .23		
Language type				5.51	.07
Receptive	4	.49***	.20, .69		
Expressive	21	.18***	.08, .27		
General language	8	.13**	.03, .22		
Language method				0.11	.99
Questionnaire	3	.20*	.01, .38		
Observation	13	.21**	.06, .35		
Direct assessment	8	.18**	.07, .29		
Combination	3	.21*	.02, .38		
Language informant				.06	.99
Parent	9	.20*	.01, .38		
Trained coder	14	.19*	.04, .32		
Experimenter	7	.21**	.09, .32		
Combination	3	.21*	.02, .38		
Dissemination medium				0.64	.43
Dissertation	9	.15***	.06, .24		
Publication	24	.21***	.10, .30		
Study design				5.35	.02
Cross-sectional	16	.12**	.04, .20		
Longitudinal	17	.29***	.17, .39		
Continuous moderator	<i>k</i>	<i>b</i>	95% CI	Z-value	<i>p</i> -value
Child age at parent measure	33	−.001	−.008, .006	−0.29	.78
Child age at language measure	33	.001	−.005, .008	0.37	.72
Percent of females in sample	33	.006	−.004, .017	1.22	.23
Percent of male parents	33	−.001	−.005, .004	−0.29	.77
Observation length ^a	31	.000	−.001, .001	0.44	.66

^aThe Hart and Risley (1995) study was removed from this analysis as it was an outlier (3 + *SDs*) in terms of observation length.**p* < .05. ***p* < .01. ****p* < .001.

these aspects of linguistic quality was driving the moderator result of age on the relation between linguistic input quality and child language. When assessed individually, neither of these associations were moderated by age. As such, in our analyses, age was only demonstrated to be a significant moderator of the global measure of linguistic input quality and child language outcomes. This evidence may support the notion that what constitutes

optimal input may change in accordance with children's maturing language abilities (Jones & Rowland, 2017).

Additional Moderators of Associations

Neither child nor parent gender moderated the association between linguistic input and child language, however, the context in which parent

language was observed did moderate the association between quality and quantity of linguistic input and child language. Specifically, measurements taken during naturalistic observations yielded higher effect sizes than those taken during free play tasks or any combination of tasks. Free play tasks are typically shorter, conducted in the research laboratory, and may limit the input somewhat due to a focus on the toys available to the dyad. In contrast, naturalistic observations, typically conducted in the comfort of a dyad's home, for longer durations, may yield more natural conversation and provide a context for more ample and sophisticated vocabulary compared to free-play (Pan, Perlmann, & Snow, 2000; Snow & Beals, 2006). This contextual finding will be an important consideration for future research seeking to determine the most optimal context for observing language interactions (Beals & Tabors, 1993; Bornstein et al., 1999).

Further demonstrating the importance of measurement context in characterizing the relations between parent and child speech, our analyses also showed that the length of the observation significantly moderated the association between the quality of parental linguistic input and child language, such that longer observation periods were associated with larger effect sizes. In line with the arguments above, it may be that longer observational periods yield a more representative sampling of language, which is consistent with the notion that researchers should attempt to observe children and parents in the context most representative of their day-to-day lives. In sum, future language research using observational methods may benefit from assessing parent and child language naturalistically and for longer periods if they wish to capture the most representative sample of parental linguistic quality.

Next, we also found that effect sizes for the association between quantity of input and child language were stronger when examined longitudinally versus cross-sectionally. This finding may reflect the notion that the role of quantity of linguistic input may accumulate over time (i.e., an accumulating word gap; Hart & Risley, 1995). That is, it is possible that differences in language ability and input that exist at one point in time are continually and reciprocally compounded throughout development (consistent with a transactional framework; e.g. Barnett, Gustafsson, Deng, Mills-Koonce, & Cox, 2012). For example, a child with ample linguistic input and precocious language skills will likely elicit increased parental input over time, further strengthening their language skills.

Amid the current debate surrounding the "30 million word gap" (Hart & Risley, 1995), our finding that the relation between parental linguistic input and child language did not differ based on SES, maternal education, or other risk factors is notable. To be clear, our findings *do not* pertain directly to the presence or absence of a SES-driven word gap. Rather, these findings suggest that the association between the amount and quality of parental language and children's language is not restricted to any specific demographic group. That is, the positive relations between linguistic input and child language is seen across varied demographic groups. However, in interpreting these findings, it is also important to acknowledge the apparent sample homogeneity of the studies included on our analyses. Thirty-two of the 52 included samples (62%) were from middle to high SES groups, 12 were from diverse backgrounds (23%), and only seven (13%) were of lower SES (one sample did not provide sufficient information to determine participant SES). Moreover, only one study was characterized as having participants that were ethnically diverse. Further insight into how SES may impact the associations between parental input and child language skills will come from future research including participants from a broader range of socioeconomic and cultural backgrounds.

Finally, studies included in the current meta-analysis examined child language through observation, structured assessments, and parent reports. Effect sizes were similar across these assessment methods. Relatedly, the location of the parent observation of linguistic input was not found to moderate associations. This result suggests that useful information can be obtained across observation locations, which is in line with previous findings demonstrating that child language does not differ when assessed in the home versus the research lab (Bornstein, Haynes, Painter, & Genevro, 2000). That said, an important caveat is that the observation context and length of observation did emerge as important measurement factors and should therefore be considered in future research when studying parent and child speech.

Limitations and Future Directions

The results of the current meta-analysis must be considered in light of several limitations. First, although examined separately, quantity and quality of parental linguistic input are inextricably linked. Jones and Rowland (2017) noted, for example, that

parents who use a higher quantity of language also have more diverse language. The high degree of multicollinearity between quantity and quality of language has previously challenged researchers examining both types of input in the same statistical models (e.g., Hoff & Naigles, 2002; Huttenlocher, Waterfall, Vasilyeva, Vevea, & Hedges, 2010). Second, the statistics included in the present analyses were correlational. As such, it is impossible to infer causal meaning from the present findings. We sought to minimize issues of reverse causality by selecting associations in which the parent language measurement temporally preceded the child language measure when they were available. Even with these precautions, we note that we cannot conclude a causal association between parental input and child language skills.

Third, as noted previously, parent language does not fully capture the complexity of the linguistic environment to which children are exposed (e.g., overheard speech; Sperry et al., 2019a, 2019b), and as such, our analyses examined only two environmental features relevant to children's language development in English. Additionally, we selected a specific definition of parental linguistic input that included the quality and quantity of the words spoken by parents to the child, with quality exclusively referring to vocabulary diversity and syntactic complexity. This was a strength of the analyses in that it enabled us to examine a uniform construct of linguistic input, and we also made efforts to include as much nuance as possible by exploring diversity and complexity separately in sub-analyses. This definition, however, could also be viewed as a limitation as it does not fully capture the complexity of linguistic input, particularly with respect to input quality (Rowe & Snow, 2020). Furthermore, elucidating additional features of the linguistic environment that are linked to children's language is necessary for a comprehensive understanding of children's language-learning. Recent meta-analytic evidence suggests that parental sensitive responsiveness is one such modifiable factor, as it was shown to have a moderate association with children's language skills (Madigan et al., 2019). In line with the importance of sensitive responsiveness, the ability of a parent to tailor their input to the developmental level, interest, and needs of the child should also be considered. Finally, other key candidates for future inquiry include other aspects of the environment such as conversational turn taking (Romeo et al., 2018) and verbal responsivity (Tamis-LeMonda, Kuchirko, & Song, 2014).

A fourth limitation arises from the fact that the number of studies that included fathers in their

analyses was relatively small ($k = 15$, with percentage of fathers in the sample ranging from 2% to 50%, $M = 30.48\%$). Paternal linguistic input is a burgeoning area in the literature (e.g., Baker et al., 2015; Leech et al., 2013), and further research in this area is undoubtedly needed in order to fully understand the association between linguistic input and child language. Additionally, research should continue to examine the potential influence of other family members (e.g., siblings) and family dynamics on child speech (e.g., Prime, Pauker, Plamondon, Perlman, & Jenkins, 2014; Sperry et al., 2019a). Additionally, the studies included focused on English-speaking children (consistent with other meta-analyses on child language outcomes; Madigan et al., 2019; Milligan, Astington, & Dack, 2007) and many of the samples included in this meta-analysis consisted primarily of Caucasian, middle- to upper-SES participants. In order to increase the generalizability of findings, efforts to include more diverse samples in terms of SES, minority status, languages spoken, and cultural factors should continue. Additionally, the current analyses were restricted to typically developing children. As such, further research is needed in order to examine these associations in children with known risks (e.g., Autism Spectrum Disorder, Language Disorder).

Finally, as noted previously, despite efforts to include all relevant studies, several studies were excluded due to lack of returned responses to our requests for correlational values, or inaccessibility of archival data (see Table S5). Although this issue is common to all meta-analyses, it is worth noting that conclusions derived in this meta-analysis can only be based on the studies available for inclusion. Fortunately, we attained a large number of studies with extractable data and therefore, we consider our meta-analysis sufficiently comprehensive.

Conclusions

Both the quantity and quality of parental linguistic input is associated with child language skills, with quality more strongly associated with child language skills than quantity, especially as children age. Results from the current study can be used to inform future language research, specifically suggesting that observational measures would benefit from approximating families' natural environments and longer observational periods to gather the most representative data possible. Notably, the pooled effect sizes derived herein are similar in magnitude to a meta-analysis on the association between maternal sensitive responsiveness and child

language across 36 studies ($r = .27$; 95% CI [.21, .33]; Madigan et al., 2019). Together, these findings speak to the important role of examining environmental contributions to children's language development, and identify important targets for intervention. Future research should devote concerted attention to elucidating how input from other caregivers and from overheard speech work in concert with parental linguistic input to shape children's language skills. Moreover, concerted attention to elucidating the relative importance of different parenting behaviors such as sensitivity, verbal responsiveness, joint attention, and linguistic input (Wade, Jenkins, Venkadasalam, Binnoo-Erez, & Ganea, 2018) are needed in order to further understand the role of early environmental experiences in children's developing language abilities.

References

- Agostin, T. M., & Bain, S. K. (1997). Predicting early school success with developmental and social skills screeners. *Psychology in the Schools*, 34, 219–228. [https://doi.org/10.1002/\(SICI\)1520-6807\(199707\)34:3<219:AID-PITS4>3.0.CO;2-J](https://doi.org/10.1002/(SICI)1520-6807(199707)34:3<219:AID-PITS4>3.0.CO;2-J)
- Akhtar, N., Jipson, J., & Callanan, M. A. (2001). Learning words through overhearing. *Child Development*, 72, 416–430. <https://doi.org/10.1111/1467-8624.00287>
- Akhtar, N., Tolins, J., & Tree, J. E. F. (2019). Next steps. In J. Horts & J. von Koss Torkildsen (Eds.), *International handbook of language acquisition* (pp. 427–441). London, UK: Routledge.
- Baker, C. E., Vernon-Feagans, L.; Family Life Project Investigators. (2015). Fathers' language input during shared book activities: Links to children's kindergarten achievement. *Journal of Applied Developmental Psychology*, 36, 53–59. <https://doi.org/10.1016/j.appdev.2014.11.009>
- Barnett, M. A., Gustafsson, H., Deng, M., Mills-Koonce, W. R., & Cox, M. (2012). Bidirectional associations among sensitive parenting, language development, and social competence. *Infant and Child Development*, 21, 374–393. <https://doi.org/10.1002/icd.1750>
- Beals, D. E., & Tabors, P. O. (1993). *Arboretum, bureaucratic, and carbohydrates: Preschoolers' exposure to rare vocabulary at home*. Paper presented at the Biennial Meeting of the Society for Research in Child Development, New Orleans, LA, March 25–28, 1993.
- Bernstein Ratner, N. (1988). Patterns of parental vocabulary selection in speech to very young children. *Journal of Child Language*, 15, 481–492. <https://doi.org/10.1017/S0305000900012514>
- Bingham, G. E., Kwon, K. A., & Jeon, H. J. (2013). Examining relations among mothers', fathers', and children's language use in a dyadic and triadic context. *Early Child Development and Care*, 183, 394–414. <https://doi.org/10.1080/03004430.2012.711590>
- Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. West Sussex, UK: Wiley.
- Borenstein, M., Rothstein, D., & Cohen, J. (2005). *Comprehensive meta-analysis: A computer program for research synthesis [Computer software]*. Englewood, NJ: Biostat.
- Bornstein, M. H., & Haynes, O. M. (1998). Vocabulary competence in early childhood: Measurement, latent construct, and predictive validity. *Child Development*, 69, 654–671. <https://doi.org/10.1111/j.1467-8624.1998.tb06235.x>
- Bornstein, M. H., Haynes, O. M., Painter, K. M., & Genevro, J. L. (2000). Child language with mother and with stranger at home and in the laboratory: A methodological study. *Journal of Child Language*, 27, 407–420. <https://doi.org/10.1017/S0305000900004165>
- Bornstein, M. H., Tamis-LeMonda, C. S., & Haynes, O. M. (1999). First words in the second year: Continuity, stability, and models of concurrent and predictive correspondence in vocabulary and verbal responsiveness across age and context. *Infant Behavior and Development*, 22, 65–85. [https://doi.org/10.1016/S0163-6383\(99\)80006-X](https://doi.org/10.1016/S0163-6383(99)80006-X)
- Crystal, D. (2008). *Language library: A dictionary of linguistics and phonetics* (6th ed.). Hoboken, NJ: Wiley.
- Duval, S., & Tweedie, R. (2000). Trim and fill: A simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics*, 56, 455–463. <https://doi.org/10.1111/j.0006-341X.2000.00455.x>
- Fearon, R. P., Bakermans-Kranenburg, M. J., van IJzendoorn, M. H., Lapsley, A.-M., & Roisman, G. I. (2010). The significance of insecure attachment and disorganization in the development of children's externalizing behavior: a meta-analytic study. *Child Development*, 81, 435–456. <https://doi.org/10.1111/j.1467-8624.2009.01405.x>
- Feldman, H. M., Dale, P. S., Campbell, T. F., Colborn, D. K., Kurs-Lasky, M., Rockette, H. E., & Paradise, J. L. (2005). Concurrent and predictive validity of parent reports of child language at ages 2 and 3 years. *Child Development*, 76, 856–868. <https://doi.org/10.1111/j.1467-8624.2005.00882.x>
- Fenson, L., Dale, P. S., Reznick, J. S., Bates, E., Thal, D. J., Pethick, S. J., Stiles, J. (1994). Variability in early communicative development. *Monographs of the Society for Research in Child Development*, 59(5), i–185. <https://doi.org/10.2307/1166093>
- Funder, D. C., & Ozer, D. J. (2019). Evaluating effect size in psychological research: Sense and nonsense. *Advances in Methods and Practices in Psychological Science*, 2, 156–168. <https://doi.org/10.1177/2515245919847202>
- Goldstein, H., & Healy, M. J. (1995). The graphical presentation of a collection of means. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 158, 175–177. <https://doi.org/10.2307/2983411>

- Golinkoff, R. M., Hoff, E., Rowe, M. L., Tamis-LeMonda, C. S., & Hirsh-Pasek, K. (2019). Language matters: Denying the existence of the 30-million-word gap has serious consequences. *Child Development, 90*, 985–992. <https://doi.org/10.1111/cdev.13128>
- Hart, B., & Risley, T. R. (1995). *Meaningful differences in the everyday experience of young American children*. Baltimore, MD: Paul H Brookes.
- Hedges, L., & Olkin, I. (1985). *Statistical models for meta-analysis*. New York, NY: Academic Press.
- Hentges, R., Devereux, C., Graham, S. A., & Madigan, S. (in press). Child language difficulties and internalizing and externalizing symptoms: A meta-analysis. *Child Development*.
- Herbert-Myers, H. G., Guttentag, C. L., Swank, P. R., Smith, K. E., & Landry, S. H. (2006). The importance of language, social, and behavioral skills across early and later childhood as predictors of social competence with peers. *Applied Developmental Science, 10*, 174–187. https://doi.org/10.1207/s1532480xads1004_2
- Hoff, E. (2003). The specificity of environmental influence: Socioeconomic status affects early vocabulary development via maternal speech. *Child Development, 74*, 1368–1378. <https://doi.org/10.1111/1467-8624.00612>
- Hoff, E. (2006). How social contexts support and shape language development. *Developmental Review, 26*, 55–88. <https://doi.org/10.1016/j.dr.2005.11.002>
- Hoff, E. (2013). Interpreting the early language trajectories of children from low-SES and language minority homes: Implications for closing achievement gaps. *Developmental Psychology, 49*, 4. <https://doi.org/10.1037/a0027238>
- Hoff, E., & Naigles, L. (2002). How children use input to acquire a lexicon. *Child Development, 73*, 418–433. <https://doi.org/10.1111/1467-8624.00415>
- Hsu, N., Hadley, P. A., & Rispoli, M. (2017). Diversity matters: Parent input predicts toddler verb production. *Journal of Child Language, 44*, 63–86. <https://doi.org/10.1017/S030500091500069>
- Huttenlocher, J., Haight, W., Bryk, A., Seltzer, M., & Lyons, T. (1991). Early vocabulary growth: Relation to language input and gender. *Developmental Psychology, 27*, 236. <https://doi.org/10.1037/0012-1649.27.2.236>
- Huttenlocher, J., Vasilyeva, M., Waterfall, H. R., Vevea, J. L., & Hedges, L. V. (2007). The varieties of speech to young children. *Developmental Psychology, 43*, 1062. <https://doi.org/10.1037/0012-1649.43.5.1062>
- Huttenlocher, J., Waterfall, H., Vasilyeva, M., Vevea, J., & Hedges, L. V. (2010). Sources of variability in children's language growth. *Cognitive Psychology, 61*, 343–365. <https://doi.org/10.1016/j.cogpsych.2010.08.002>
- Johnson, C. J., Beitchman, J. H., & Brownlie, E. B. (2010). Twenty-year follow-up of children with and without speech-language impairments: Family, educational, occupational, and quality of life outcomes. *American Journal of Speech-Language Pathology, 19*, 51–65. [https://doi.org/10.1044/1058-0360\(2009/08-0083\)](https://doi.org/10.1044/1058-0360(2009/08-0083))
- Jones, G., & Rowland, C. F. (2017). Diversity not quantity in caregiver speech: Using computational modeling to isolate the effects of the quantity and the diversity of the input on vocabulary growth. *Cognitive Psychology, 98*, 1–21. <https://doi.org/10.1016/j.cogpsych.2017.07.002>
- Julious, S. A. (2004). Using confidence intervals around individual means to assess statistical significance between two means. *Pharmaceutical Statistics, 3*, 217–222. <https://doi.org/10.1002/pst.126>
- Laks, D. R., Beckwith, L., Cohen, S. E., & Cohen, S. E. (1990). Mothers' use of personal pronouns when talking with toddlers. *The Journal of Genetic Psychology, 151*, 25–32. <https://doi.org/10.1080/00221325.1990.9914641>
- Law, J., Garrett, Z., & Nye, C. (2004). The efficacy of treatment for children with developmental speech and language delay/disorder: A meta-analysis. *Journal of Speech, Language, and Hearing Research, 47*, 924–943. [https://doi.org/10.1044/1092-4388\(2004/069\)](https://doi.org/10.1044/1092-4388(2004/069))
- Leaper, C., Anderson, K. J., & Sanders, P. (1998). Moderators of gender effects on parents' talk to their children: A meta-analysis. *Developmental Psychology, 34*, 3. <https://doi.org/10.1037/0012-1649.34.1.3>
- Leaper, C., & Smith, T. E. (2004). A meta-analytic review of gender variations in children's language use: Talkativeness, affiliative speech, and assertive speech. *Developmental Psychology, 40*, 993. <https://doi.org/10.1037/0012-1649.40.6.993>
- Leech, K. A., Salo, V. C., Rowe, M. L., & Cabrera, N. J. (2013). Father input and child vocabulary development: The importance of wh-questions and clarification requests. *Seminars in Speech and Language, 34*, 249–259. <https://doi.org/10.1055/s-0033-1353445>
- Madigan, S., Prime, H., Graham, S., Rodrigues, M., Anderson, N., Khoury, J., & Jenkins, J. (2019). Parenting behavior and child language: A meta-analysis. *Pediatrics, 154*, e20183556. <https://doi.org/10.1542/peds.2018-3556>
- Milligan, K., Astington, J. W., & Dack, L. A. (2007). Language and theory of mind: Meta-analysis of the relation between language ability and false-belief understanding. *Child Development, 78*, 622–646. <https://doi.org/10.1111/j.1467-8624.2007.01018.x>
- Mol, S. E., & Neuman, S. B. (2014). Sharing information books with kindergartners: The role of parents' extra-textual talk and socioeconomic status. *Early Childhood Research Quarterly, 29*, 399–410. <https://doi.org/10.1016/j.ecresq.2014.04.001>
- Pan, B. A., Perlmann, R. Y., & Snow, C. E. (2000). Food for thought: Dinner table as a context for observing parent-child discourse. In L. Menn & B. Ratner (Eds.), *Methods for studying language production* (pp. 205–224). Mahwah, NJ: Erlbaum.
- Prime, H., Pauker, S., Plamondon, A., Perlman, M., & Jenkins, J. (2014). Sibship size, sibling cognitive sensitivity, and children's receptive vocabulary. *Pediatrics, 133*, e394–e401. <https://doi.org/10.1542/peds.2012-2874>
- Purpura, D. J. (2019). Language clearly matters; Methods matter too. *Child Development, 90*, 1839–1846. <https://doi.org/10.1111/cdev.13327>

- Quinn, S., Donnelly, S., & Kidd, E. (2018). The relationship between symbolic play and language acquisition: A meta-analytic review. *Developmental Review, 49*, 121–135. <https://doi.org/10.1016/j.dr.2018.05.005>
- Roben, C. K., Cole, P. M., & Armstrong, L. M. (2013). Longitudinal relations among language skills, anger expression, and regulatory strategies in early childhood. *Child Development, 84*, 891–905. <https://doi.org/10.1111/cdev.12027>
- Roberts, M. Y., Curtis, P. R., Sone, B. J., & Hampton, L. H. (2019). Association of parent training with child language development: A systematic review and meta-analysis. *JAMA Pediatrics, 2019*.1197. <https://doi.org/10.1001/jamapediatrics.2019.1197>
- Roberts, M. Y., & Kaiser, A. P. (2011). The effectiveness of parent-implemented language interventions: A meta-analysis. *American Journal of Speech-Language Pathology, 20*, 180–199. [https://doi.org/10.1044/1058-0360\(2011/10-0055\)](https://doi.org/10.1044/1058-0360(2011/10-0055))
- Romeo, R. R., Segaran, J., Leonard, J. A., Robinson, S. T., West, M. R., Mackey, A. P., . . . Gabrieli, J. D. E. (2018). Language exposure relates to structural neural connectivity in childhood. *Journal of Neuroscience, 38*, 7870–7877. <https://doi.org/10.1523/JNEUROSCI.0484-18.2018>
- Rosenthal, R. (1995). Writing meta-analytic reviews. *Psychological Bulletin, 118*, 183. <https://doi.org/10.1037/0033-2909.118.2.183>
- Rowe, M. L. (2012). A longitudinal investigation of the role of quantity and quality of child-directed speech in vocabulary development. *Child Development, 83*, 1762–1774. <https://doi.org/10.1111/j.1467-8624.2012.01805.x>
- Rowe, M. L., & Snow, C. E. (2020). Analyzing input quality along three dimensions: Interactive, linguistic, and conceptual. *Journal of Child Language, 1*, 17. <https://doi.org/10.1017/S0305000919000655>
- Snow, C. E., & Beals, D. E. (2006). Mealtime talk that supports literacy development. *New Directions for Child and Adolescent Development, 2006*, 51–66. <https://doi.org/10.1002/cd.155>
- Sperry, D. E., Sperry, L. L., & Miller, P. J. (2019a). Reexamining the verbal environments of children from different socioeconomic backgrounds. *Child Development, 90*, 1303–1318. <https://doi.org/10.1111/cdev.13072>
- Sperry, D. E., Sperry, L. L., & Miller, P. J. (2019b). Language does matter: But there is more to language than vocabulary and directed speech. *Child Development, 90*, 993–997. <https://doi.org/10.1111/cdev.13125>
- Stromswold, K. (2001). The heritability of language: A review and metaanalysis of twin, adoption, and linkage studies. *Language, 647*–723. <https://doi.org/10.1353/lan.2001.0247>
- Tamis-LeMonda, C. S., Kuchirko, Y., & Song, L. (2014). Why is infant language learning facilitated by parents' contingent speech? *Current Directions in Psychological Science, 23*, 121–126. <https://doi.org/10.1177/0963721414522813>
- Thompson, S. G., & Higgins, J. P. (2002). How should meta-regression analyses be undertaken and interpreted? *Statistics in Medicine, 21*, 1559–1573. <https://doi.org/10.1002/sim.1187>
- Turkheimer, E., Haley, A., Waldron, M., d'Onofrio, B., & Gottesman, I. I. (2003). Socioeconomic status modifies heritability of IQ in young children. *Psychological Science, 14*, 623–628. <https://doi.org/10.1046/j.0956-7976.2003.psci.1475.x>
- Vernon-Feagans, L., Bratsch-Hines, M., Reynolds, E., & Willoughby, M. (2019). How early maternal language input varies by race and education and predicts later child language. *Child Development, 91*, 1098–1115. <https://doi.org/10.1111/cdev.13281>
- Wade, M., Browne, D. T., Madigan, S., Plamondon, A., & Jenkins, J. M. (2014). Normal birth weight variation and children's neuropsychological functioning: Links between language, executive functioning, and theory of mind. *Journal of the International Neuropsychological Society, 20*, 909–919. <https://doi.org/10.1017/S1355617714000745>
- Wade, M., Jenkins, J. M., Venkadasalam, V. P., Binnoon-Erez, N., & Ganea, P. A. (2018). The role of maternal responsiveness and linguistic input in pre-academic skill development: A longitudinal analysis of pathways. *Cognitive Development, 45*, 125–140. <https://doi.org/10.1016/j.cogdev.2018.01.005>

Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's website:

Figure S1. Funnel Plot of the Studies Examining the Association Between Quality of Parental Linguistic Input and Child Language.

Figure S2. Funnel Plot of the Studies Examining the Association Between Quantity of Parental Linguistic Input and Child Language

Table S1. Search Strategy

Table S2. Inclusion Criteria

Table S3. Demographic Characteristics of Studies Included in the Meta-Analyses on Parental Linguistic Input and Child Language

Table S4. Characteristics of Studies Included in the Meta-Analyses on Parental Linguistic Input and Child Language

Table S5. Author Contacts to Obtain Raw Data